

# Manhwa Recap Studio v3

## OCR Accuracy & Garble Fix Guide

Targeted research for zainrana558/manhwa-recap-studio-v3

Why current RapidOCR setup produces garbles · Accuracy-first tools · Detection · Post-correction · Action plan

Compiled 2026-09-10 · No-GPU / CPU-first · Quality over pure speed

### 1. Why Your Current Setup Produces a Lot of Garble

Your pipeline uses RapidOCR (PP-OCRv5/v6 ONNX) as primary, PaddleOCR PP-OCRv4 as fallback, and VLMs as last resort, with YOLO panel detection. This is a reasonable practical stack, but general-purpose OCR engines were trained on clean horizontal print. Manhwa/webtoon text breaks those assumptions.

#### Primary root causes of garble (ranked by impact)

- **1. Full-page or loose-crop OCR** — Background art, gradients, screentones and panel borders are read as characters. This is the #1 source of insertions and nonsense strings.
- **2. Generic detectors** — Standard YOLO/panel models miss irregular bubbles, free text outside bubbles, SFX, and extreme aspect-ratio webtoon strips.
- **3. Wrong language model pack** — Using Chinese (ch) recognition models on English dialogue drops spaces and confuses Latin characters (documented RapidOCR issue: CER ~0.136 with ch vs ~0.050 with en models on the same English scans).
- **4. Stylized / decorative fonts** — Hand-drawn lettering, bold SFX, and mixed full-width/half-width characters are outside the training distribution of generic PP-OCR.
- **5. No English post-correction** — Classic confusions (l↔I, 0↔O, pipe→I, missing/extra spaces) remain unfixed and pollute the recap script.
- **6. Vertical / mixed-direction text** — Especially on Korean manhwa and Japanese pages; general detectors and recognizers degrade.

Research consensus (COMICS Text+, Manga109 evaluations, RapidOCR discussions, manga-OCR papers): layout-aware detection + bubble crops + domain-adapted recognition + English pattern correction is required. Generic engines alone remain inadequate for complex comic layouts.

### 2. Current Stack vs Accuracy-First Target

| Stage                | Your current             | Accuracy-first target                                         | Why                                                             |
|----------------------|--------------------------|---------------------------------------------------------------|-----------------------------------------------------------------|
| Panel / strip        | YOLO panel               | Keep + gutter-aware long-strip split                          | Long vertical webtoons need gutters first                       |
| Bubble / text detect | Generic YOLO             | ogkalu RT-DETR or YOLOv8m (webtoon-trained)                   | Classes: bubble, text_bubble, text_free; handles extreme ratios |
| OCR primary          | RapidOCR (often ch pack) | RapidOCR with <b>en</b> (or lang-specific) pack on crops only | Spaces + Latin accuracy jump                                    |
| OCR specialist       | —                        | Baberu OCR (ONNX) on bubble crops                             | Bubble-trained; competitive EN/JA/ZH                            |
| OCR hard fallback    | Paddle v4 / VLM          | Hayai LiteRT (KR) or quantized PaddleOCR-VL-For-Manga         | Higher exact-match on stylized text                             |
| Post-correct         | None / minimal           | Zentrix-style English patterns + series dictionary            | Fixes residual I/I/O/O/spacing                                  |

### 3. Accuracy-Ranked Tools (CPU-First)

#### 3.1 Detection (do this before any OCR)

**ogkalu/comic-text-and-bubble-detector** (RT-DETR-v2, ~43 M, ONNX available) — fine-tuned on ~11 k manga + webtoon + manhwa + Western images. Classes: bubble / text\_bubble / text\_free. Tall webtoons split vertically during training. **ogkalu YOLOv8m speech-bubble** — trained at 1024, explicitly handles extreme Korean webtoon aspect ratios. YOLO26-Nano / Kiuyha bubble models — very light, high mAP, used in browser pipelines. dmMaze/comic-text-detector and CTD remain solid alternatives for line-level masks.

**Always OCR only the cropped regions. This single change usually cuts more garbles than swapping the recognizer.**

#### 3.2 Recognition (quality order for your pipeline)

| Tool                   | Size                         | Strength for manhwa recaps                                                                             | CPU notes                     |
|------------------------|------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------|
| Baberu OCR             | 115 M · 121–242 MB ONNX      | Bubble specialist; EN per-bubble ICER & exact-match competitive with 0.9 B VL; JA/ZH/EN one checkpoint | onnxruntime only — fast       |
| Hayai OCR v2.1         | Quantizable + LiteRT         | Multi-line, vertical, SFX, Korean; strong mixed benchmarks                                             | LiteRT = no PyTorch           |
| RapidOCR en pack       | ~15–50 MB                    | Best default bulk English when used on crops; use LangRec.EN                                           | Already in your stack         |
| PaddleOCR-VL-For-Manga | ~0.5–0.9 B (quantized ports) | Full-sentence accuracy 27%→~70% on manga crops in fine-tune report                                     | Heavier; use as hard fallback |
| manga-ocr-torchless    | ~400 MB ONNX                 | Classic multi-line JA bubble OCR, Torch-free                                                           | Good JA specialist            |

**Important RapidOCR finding: Chinese recognition models on English text lose spaces and inflate CER. Always select the English (or language-matched) recognition pack for English dialogue.**

#### 3.3 English post-OCR correction

**ZentrixLabs.OcrCorrection** (~865 patterns): pipe→I, capital l↔lowercase l, spacing, apostrophes, multi-pass. Validated on 10 k+ subtitle lines with zero remaining OCR errors and zero false positives in their tests. COMICS Text+ style rules (frequency vocab + first/last-letter + single-char cleanup) also help. Maintain a small series dictionary of character names and recurring slang for edit-distance fallback.

### 4. Concrete Upgrade Plan for manhwa-recap-studio-v3

Ordered by impact on garble reduction. All steps remain CPU-compatible with your Oracle production mode.

**Step 1 — Detection first (highest ROI)**

Keep gutter/panel split for long strips. Add ogkalu RT-DETR (or YOLOv8m) for bubble + text\_free. Crop tightly. Reject tiny/noisy boxes by area ratio.

**Step 2 — Language-correct RapidOCR**

Force English recognition pack (Rec.lang\_type = EN, clear any pinned ch model path). Same for Korean pack on raw manhwa. Run only on crops.

**Step 3 — Specialist cascade**

Crop → RapidOCR (lang pack) → if low confidence or short/garbled → Baberu ONNX → if still bad → Hayai LiteRT (KR/JA) or quantized PaddleOCR-VL-For-Manga → VLM last.

**Step 4 — English post-correct**

Zentrix-style pattern pass + series name dictionary on every English string before script generation.

**Step 5 — Optional light preprocess on crops**

Mild contrast / adaptive threshold / denoise on the bubble crop only (not full page). Avoid aggressive binarization that destroys stylized strokes.

**Expected effect: fewer insertions from art, correct spaces on English, lower character confusions, cleaner input to the LLM script stage → less nonsense narration.**

5. Recommended Accuracy Stacks by Content Type

| Content                      | Stack                                           | Notes                              |
|------------------------------|-------------------------------------------------|------------------------------------|
| English webtoon / translated | ogkalu detect → RapidOCR en → Baberu → Zentrix  | Primary path for most recaps       |
| Raw Korean manhwa            | ogkalu detect → Hayai or RapidOCR ko → Baberu   | Vertical + Hangul                  |
| Mixed / hard stylized        | ogkalu → Baberu primary → VL-For-Manga fallback | Promote specialist earlier         |
| Bulk chapters (CPU)          | Same cascade; batch crops; confidence gate      | Fits existing service architecture |

6. Key Links (Quick Reference)

**ogkalu bubble+text detector:** [huggingface.co/ogkalu/comic-text-and-bubble-detector](https://huggingface.co/ogkalu/comic-text-and-bubble-detector)  
**ogkalu YOLOv8m speech-bubble:** [huggingface.co/ogkalu/comic-speech-bubble-detector-yolov8m](https://huggingface.co/ogkalu/comic-speech-bubble-detector-yolov8m)  
**Baberu OCR:** [huggingface.co/genshi-ai-daichi/baberu-ocr](https://huggingface.co/genshi-ai-daichi/baberu-ocr)  
**Hayai OCR:** [github.com/NopeNopeGuy/hayai-ocr](https://github.com/NopeNopeGuy/hayai-ocr)  
**RapidOCR (use en pack):** [github.com/RapidAI/RapidOCR](https://github.com/RapidAI/RapidOCR)  
**PaddleOCR-VL-For-Manga:** [huggingface.co/jzhang533/PaddleOCR-VL-For-Manga](https://huggingface.co/jzhang533/PaddleOCR-VL-For-Manga)  
**ZentrixLabs.OcrCorrection:** [github.com/ZentrixLabs/ZentrixLabs.OcrCorrection](https://github.com/ZentrixLabs/ZentrixLabs.OcrCorrection)  
**Your repo:** [github.com/zainrana558/manhwa-recap-studio-v3](https://github.com/zainrana558/manhwa-recap-studio-v3)

7. Bottom Line

Garble is not primarily a “need a faster model” problem — it is a **detection + crop + language-pack + domain model + English correction** problem. Your RapidOCR service is fine as a fast primary **once it only sees tight bubble crops and uses the correct language pack**. Add ogkalu (or equivalent webtoon-trained) detection, Baberu as specialist, and Zentrix-style English post-correction. Keep VLMs as last resort only. Measure on a small set of your own chapter pages (CER / exact-match on dialogue) before and after to confirm the lift.

Consolidated from deep research on comic OCR benchmarks, RapidOCR language-pack issues, Baberu/Hayai/PaddleOCR-VL-For-Manga results, ogkalu detectors, and English post-correction libraries. All recommended tools run offline on CPU after model download.